Tutorial Questions 04 – The derivative Dr Killian O'Brien
6G4Z9019 Mathematical Methods 09/11/26

Continuity

1. **Quotient case of theorem 4.3:** Suppose $g(x)$ is non-zero in some interval of the domain containing the point a and that g and another function f are continuous at a. Prove that the quotient function f/g defined by $(f/g)(x) = f(x)/g(x)$ is continuous at $x = a$. You should follow the principles used for the similar result of the quotient case of the algebra of limits theorem for quotients.

2. Use the formal definition of continuity to show that the function $f(t) = t^2$ is continuous at every point in its domain.

The derivative

3. Can you confidently quote the limit definition of the derivative from memory? Can you confidently quote it using different notation than the usual f and x? For instance what is the definition for the derivative of a function λ of a real variable y at a point c of its domain?

4. Use the limit definition of the derivative to obtain the derivative of the function defined by $g(t) = t^5$.

5. Generalise the previous question to show that for each $n > 0$ if $h(t) = t^n$ then $h'(t) = nt^{n-1}$.

6. Use the limit definition of the derivative to obtain the derivative of the function defined by $h(t) = t^{-2}$.

7. Generalise the previous question to show that for each $n > 0$ if $h(t) = t^{-n}$ then $h'(t) = -nt^{-n-1}$.

8. What does the product rule look like for products of three or more functions? That is, suppose functions f, g, h are all differentiable at a point $x = a$. Find an expression for the derivative at a of the product function fgh defined by $(fgh)(x) = f(x)g(x)h(x)$, Your expression will feature the derivatives $f'(a), g'(a), h'(a)$ and the values $f(a), g(a), h(a)$. What about a product of four functions? Or for a general product of n functions, i.e. the function defined by

$$(f_1 f_2 \dots f_n)(x) = f_1(x) f_2(x) \dots f_n(x),$$

or expressed using the Pi product notation

$$\left(\prod_{i=1}^{n} f_i \right)(x) = \prod_{i=1}^{n} f_i(x).$$

9. In a similar spirit to the previous question, what does the chain rule look like for compositions of three or more functions? For instance, what is the derivative of $f \circ g \circ h$ at a point $x = a$, how does it depend on various information about the component functions f, g, h. We can assume that f, g, h are differentiable everywhere.

   Can you generalise to the case of a composition of n functions, where n could be any positive integer?

10. Use properties of the derivative and results from the table of derivatives to obtain the derivatives of the following functions.

    a) $f(x) = x^5 + 3x^4 - 5x^2 + 10$
    b) $g(x) = 3x^{1/2} - x^{3/2} + 2x^{-1/2}$
    c) $\lambda(t) = \sqrt{3t} + 3\sqrt{t}$
    d) $\mu(y) = (2 + 5y + 1/2y^2)^{1/2}$
    e) $\phi(z) = (z+1)\sqrt{z^4 - 2z + 2}$
    f) $h(x) = x/\sqrt{1 - 4x^2}$
    g) $p(x) = \sqrt{1 + \sqrt{x}}$

11. Using geometric arguments it can be shown that

$$\lim_{h \to 0} \frac{\sin(h)}{h} = 1 \text{ and } \lim_{h \to 0} \frac{1 - \cos(h)}{h} = 0.$$

    Use these results and the relation $\sin(A + B) = \cos(A)\sin(B) + \sin(A)\cos(B)$ to carry out the limit definition of the derivative and show that

$$\frac{d}{dx}\sin(x) = \cos(x) \text{ and } \frac{d}{dx}\cos(x) = -\sin(x).$$

    Many other derivatives of trigonometric functions can be obtained by combining these results with other properties of the derivative.